THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A wheelbarrow is a: 1

(a) third class lever **(b)** pulley
(c) first class lever **(d)** second class lever

2. A 2 kg body moving at 4 m/s. KE: 1

(a) 8 J **(b)** 16 J **(c)** 4 J **(d)** 32 J

3. KE = 1

(a) m g h **(b)** m v
(c) F s **(d)** (1/2) m v squared

4. Tweezers are a: 1

(a) second class lever **(b)** first class lever
(c) third class lever **(d)** pulley

5. A person holds a box without moving. Work done: 1

(a) zero **(b)** maximum **(c)** positive **(d)** negative

6. PE = 1

(a) m g h **(b)** (1/2) m v squared
(c) m v **(d)** F/s

7. A ramp is an: 1

(a) gear **(b)** inclined plane **(c)** lever **(d)** pulley

8. A 100 W bulb for 10 h uses: 1

(a) 10 kWh **(b)** 0.1 kWh **(c)** 100 kWh **(d)** 1 kWh

9. A stone at the top of a hill has: 1

(a) kinetic energy only **(b)** sound energy
(c) no energy **(d)** potential energy

10. A machine makes work: 1

(a) unnecessary **(b)** easier **(c)** heavier **(d)** impossible

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define 1 joule. 2

12. What is mechanical advantage? 2

13. Define kinetic energy. 2

14. Define 1 watt. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Give three energy conversions. 3

16. State three uses of simple machines. 3

17. State the energy change in a pendulum. 3

OR

Find KE of 4 kg at 5 m/s. 3

18. Why is no work done when we push a wall? 3

OR

Name three forms of energy. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Derive PE = mgh. 5

OR

A machine does 6000 J in 2 min. Find the power and the energy in kWh. 5

20. A 5 kg body falls from 10 m (g = 10). Find PE, KE and speed on reaching the ground. 5

OR

Explain work-energy theorem. 5

END OF QUESTION PAPER — Total: 40 Marks

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